

Hemostasis & Thrombosis

Original MCQ Reviewer with Answer Key and Rationales

Based on the concepts and topic coverage of Chapter 2. Questions, scenarios, distractors, and rationales are newly written for study use rather than copied or lightly paraphrased from the source.

2.1 Coagulation and Fibrinolytic Systems / Reagents and Methods

1. A vascular injury exposes a membrane-associated protein that binds factor VII/VIIa and begins physiologic coagulation. Which protein is responsible?

- A. Thrombomodulin
- B. Tissue factor
- C. Protein S
- D. Plasminogen

2. A patient has an isolated deficiency of factor XII. Which laboratory finding is most consistent with this defect?

- A. Shortened PT
- B. Prolonged APTT
- C. Prolonged thrombin time
- D. Low platelet count

3. Which anticoagulant is preferred when collecting blood for routine PT and APTT testing?

- A. EDTA
- B. Heparin
- C. 3.2% sodium citrate
- D. Sodium fluoride

4. A coagulation tube should normally contain what approximate ratio of blood to citrate anticoagulant?

- A. 4:1
- B. 5:1
- C. 9:1
- D. 10:1

5. Both PT and APTT are prolonged in a patient with an isolated factor X deficiency. Why?

- A. Factor X belongs only to the intrinsic pathway
- B. Factor X belongs to the common pathway
- C. Factor X activates platelets directly
- D. Factor X is required only for fibrinolysis

6. A patient is suspected of having factor XIII deficiency despite normal PT and APTT results. Why can these screening tests remain normal?

- A. Factor XIII functions after fibrin formation
- B. Factor XIII acts only on platelets
- C. Factor XIII is measured only by PT
- D. Factor XIII is not involved in hemostasis

7. Which statement best describes the role of vitamin K in coagulation?

- A. It directly converts fibrinogen to fibrin
- B. It supports carboxylation of several coagulation proteins
- C. It activates plasmin
- D. It inhibits platelet adhesion

8. What does a D-dimer assay specifically provide evidence of?

- A. Breakdown of cross-linked fibrin
- B. Platelet destruction
- C. Isolated fibrinogen synthesis
- D. Factor VIII activation

9. Which laboratory test primarily evaluates the intrinsic and common coagulation pathways?

- A. PT
- B. APTT
- C. Bleeding time
- D. Platelet count

10. Why can a severely underfilled blue-top tube produce falsely prolonged coagulation results?

- A. Too little platelet phospholipid remains
- B. Excess citrate relative to plasma binds reagent calcium
- C. The specimen becomes hemolyzed automatically
- D. Factor XIII is activated prematurely

11. Which endogenous molecule converts plasminogen toward the fibrinolytic pathway?

- A. Tissue plasminogen activator
- B. Factor XIIIa
- C. Protein S
- D. Thromboplastin

12. Which protein is the major physiologic inhibitor of free plasmin?

- A. Protein C
- B. Antithrombin
- C. Alpha-2 antiplasmin
- D. Factor V

2.2 Platelet and Vascular Disorders

13. A patient has thrombocytopenia, microangiopathic hemolytic anemia, neurologic abnormalities, and normal PT and APTT. Which disorder is most likely?

- A. DIC
- B. TTP
- C. Hemophilia A
- D. von Willebrand disease

14. Aspirin reduces platelet aggregation primarily by inhibiting which enzyme?

- A. Cyclooxygenase
- B. Plasmin
- C. Thrombin
- D. ADAMTS13

15. Normal platelet adhesion requires interaction between von Willebrand factor and which platelet receptor?

- A. GPIIb/IIIa
- B. GPIb
- C. P2Y12
- D. PAR-1

16. A patient with classic von Willebrand disease would most likely have which finding?

- A. Marked thrombocytopenia
- B. Normal platelet count
- C. Absent fibrinogen
- D. Prolonged PT only

17. Which combination is most characteristic of Bernard-Soulier syndrome?

- A. Small platelets and thrombocytosis
- B. Giant platelets and thrombocytopenia
- C. Normal platelets and factor VIII deficiency
- D. Isolated prolonged PT

18. Platelets from a patient aggregate normally with collagen, ADP, and epinephrine but respond poorly to ristocetin. Which disorder should be considered?

- A. Glanzmann thrombasthenia
- B. Bernard-Soulier syndrome
- C. Factor IX deficiency
- D. DIC

19. Which platelet aggregation pattern is most consistent with Glanzmann thrombasthenia?

- A. Normal ristocetin response with poor responses to ADP, collagen, and epinephrine
- B. Poor ristocetin response only
- C. Normal responses to all agonists
- D. Absent response only to thrombin

20. A child develops abrupt petechiae and severe thrombocytopenia several weeks after a viral illness. PT and APTT are normal. Which diagnosis fits best?

- A. Acute immune thrombocytopenia
- B. DIC
- C. Hemophilia B
- D. Factor XIII deficiency

21. Which organism is classically associated with hemolytic uremic syndrome in children?

- A. Staphylococcus aureus
- B. Escherichia coli O157:H7
- C. Treponema pallidum
- D. Clostridium difficile

22. Storage pool disorders primarily involve abnormalities of which platelet structure?

- A. Granules
- B. Nucleus
- C. GPIb only
- D. Fibrin strands

23. Acquired TTP is commonly associated with antibodies that impair which protein?

- A. ADAMTS13
- B. Factor VIII
- C. GPIIb/IIIa
- D. Protein C

24. Which endothelial product normally inhibits platelet aggregation?

- A. Thromboxane A₂
- B. Prostacyclin
- C. Fibrinogen
- D. Factor Xa

2.3 Coagulation System Disorders

25. A patient has a normal PT, prolonged APTT, normal platelet count, and normal platelet function. Which disorder is most compatible with these findings?

- A. Hemophilia A
- B. Bernard-Soulier syndrome
- C. DIC
- D. Glanzmann thrombasthenia

26. A patient has prolonged PT and APTT but a normal thrombin time. Which isolated factor deficiency is a strong possibility?

- A. Factor VIII
- B. Factor IX
- C. Factor X
- D. Factor XIII

27. A prolonged PT normalizes after parenteral vitamin K administration. Which clinical problem could explain the original abnormality?

- A. Obstructive jaundice causing impaired vitamin K absorption
- B. Hemophilia A
- C. Bernard-Soulier syndrome
- D. Factor XIII deficiency

28. In a factor assay, patient plasma is combined with plasma specifically lacking factor VIII. What is this approach designed to determine?

- A. Platelet count
- B. Factor VIII activity
- C. Fibrin degradation
- D. VWF antigen only

29. A patient has prolonged PT and APTT, thrombocytopenia, and evidence of systemic coagulation activation. Which diagnosis is most likely?

- A. DIC
- B. Hemophilia A
- C. ITP
- D. Factor XII deficiency

30. Which clinical condition can predispose a patient to DIC?

- A. Adenocarcinoma
- B. Isolated ITP
- C. Bernard-Soulier syndrome
- D. Hemophilia B

31. Why can factor XII deficiency produce a very prolonged APTT without a typical bleeding history?

- A. Factor XII is important for the in-vitro contact pathway but is not required for normal physiologic hemostasis
- B. Factor XII affects only platelets
- C. Factor XII deficiency raises fibrinogen
- D. Factor XII activates vitamin K

32. A patient has normal PT, prolonged APTT, and abnormal ristocetin-dependent platelet testing. Which disorder best explains the combination?

- A. von Willebrand disease
- B. Factor IX deficiency
- C. Factor VII deficiency
- D. Afibrinogenemia

33. Which manifestation is especially characteristic of severe hemophilia A?

- A. Hemarthrosis
- B. Thrombocytosis
- C. Isolated petechiae
- D. Shortened APTT

34. A patient has poor wound healing and significant bleeding despite normal PT and APTT. Which deficiency should be investigated?

- A. Factor VII
- B. Factor VIII
- C. Factor XIII
- D. Factor X

35. Which form of von Willebrand disease is the most common?

- A. Type 1
- B. Type 2A
- C. Type 2B
- D. Type 3

2.4 Inhibitors, Thrombotic Disorders, and Anticoagulant Drugs

36. What is the key relationship between heparin and antithrombin?

- A. Heparin destroys antithrombin
- B. Antithrombin acts as an important heparin cofactor
- C. Heparin converts antithrombin into fibrinogen
- D. They act only on platelets

37. A prolonged APTT fails to correct after mixing patient plasma 1:1 with normal plasma. What does this finding suggest?

- A. A coagulation factor deficiency
- B. A circulating inhibitor
- C. Thrombocytosis
- D. Isolated fibrinogen excess

38. Lupus anticoagulant primarily interferes with what type of laboratory assay?

- A. Phospholipid-dependent coagulation assays
- B. Platelet counts
- C. Hemoglobin assays
- D. Fibrinogen immunoassays only

39. Which test is commonly used to screen for lupus anticoagulant?

- A. D-dimer
- B. Dilute Russell viper venom test
- C. Platelet count
- D. Thrombin time only

40. Warfarin therapy is routinely monitored using which standardized value?

- A. INR
- B. Bleeding time
- C. D-dimer
- D. Platelet closure time

41. Activated protein C, with protein S as a cofactor, limits coagulation mainly by inactivating which cofactors?

- A. Va and VIIIa
- B. IXa and Xa
- C. VIIa and XIa
- D. I and XIIIa

42. Activated protein C resistance is most commonly associated with which inherited abnormality?

- A. Factor V Leiden
- B. Hemophilia A
- C. Factor XIII deficiency
- D. Bernard-Soulier syndrome

43. Which drug blocks the platelet P2Y₁₂ ADP receptor?

- A. Warfarin
- B. Prasugrel
- C. Heparin
- D. Urokinase

44. Which assay can be used to monitor heparin by measuring its antithrombin-dependent inhibition of factor Xa?

- A. INR
- B. Chromogenic anti-factor Xa assay
- C. Bleeding time
- D. D-dimer

45. A patient taking heparin develops a major platelet-count fall and new thrombosis. What mechanism best explains HIT?

- A. Antibody against a heparin-PF4 complex
- B. Direct vitamin K inhibition
- C. Autoantibody against factor VIII
- D. Loss of VWF

46. Which anticoagulant is appropriate for a patient with HIT who requires continued anticoagulation?

- A. Additional unfractionated heparin
- B. Argatroban
- C. Aspirin alone
- D. Vitamin K

47. Which test quantifies a factor VIII inhibitor?

- A. Bethesda assay
- B. DRVVT
- C. D-dimer
- D. PFA closure time

48. Why does factor V Leiden increase the risk of thrombosis?

- A. Factor Va resists normal inactivation by activated protein C
- B. Factor V cannot be activated
- C. Protein C is overproduced
- D. Fibrinogen is absent

49. A patient taking aspirin has normal platelet count and PT. Which test is most likely to show the drug's effect?

- A. Platelet aggregation study
- B. Factor VIII assay
- C. PT
- D. Fibrinogen antigen

2.5 Hemostasis Problem-Solving

50. A 4-year-old develops petechiae and epistaxis after a recent viral infection. PT and APTT are normal, but the platelet count is $20 \times 10^9/L$. What is the most likely diagnosis?

- A. DIC
- B. Acute ITP
- C. Hemophilia A
- D. TTP

51. A teenager has lifelong bruising and postoperative bleeding. PT is normal, APTT is prolonged, platelet count is normal, ristocetin testing is abnormal, and VWF studies are decreased. What is the most likely diagnosis?

- A. Hemophilia B
- B. von Willebrand disease
- C. Glanzmann thrombasthenia
- D. DIC

52. A severely bleeding patient has very low fibrinogen. Which blood component is especially useful for concentrated fibrinogen replacement?

- A. Cryoprecipitate
- B. Washed RBCs
- C. Platelets only
- D. Albumin

53. Five days after starting heparin, a postoperative patient develops leg thrombosis. Which routine measurement is especially important when evaluating possible HIT?

- A. Platelet count
- B. Reticulocyte count
- C. ESR
- D. Factor VIII activity

54. A postpartum patient has a prolonged APTT that corrects immediately in a mixing study but becomes prolonged after two hours at 37 C. What is the most likely explanation?

- A. Time-dependent factor VIII inhibitor
- B. Factor VII deficiency
- C. Lupus anticoagulant with immediate inhibition
- D. Thrombocytosis

55. A patient with liver disease has a markedly prolonged PT and a very high INR before an invasive biopsy. What is the safest general interpretation?

- A. The abnormal coagulation result should be addressed before proceeding
- B. PT does not matter in liver disease
- C. Proceed because platelet function is sufficient
- D. Give heparin before biopsy

56. A previously normal PT and APTT suddenly become prolonged in a preoperative sample from a patient who is not receiving anticoagulants. What should the laboratory check first?

- A. The specimen for a clot or collection problem
- B. HLA type
- C. Serum IgG
- D. Bone marrow morphology

57. A PT specimen was refrigerated for longer than recommended and now gives a shorter-than-expected PT. What preanalytical effect can explain this?

- A. Cold activation of factor VII
- B. Loss of factor XIII
- C. Platelet clumping
- D. Increased citrate concentration from evaporation

58. A patient receiving heparin shows little expected anticoagulant response. Deficiency of which physiologic protein can cause heparin resistance?

- A. Antithrombin
- B. Protein S only
- C. VWF
- D. Factor XIII

59. A patient on heparin has a platelet count that falls from $350 \times 10^9/L$ to $90 \times 10^9/L$ and develops a new leg clot. An initial platelet aggregation assay for HIT is negative. Which conclusion is most appropriate?

- A. The negative assay alone does not safely exclude HIT
- B. Heparin should automatically be increased
- C. Warfarin should immediately replace heparin without further assessment
- D. The platelet fall is insignificant

60. A patient has thrombosis, prolonged APTT, no bleeding history, and a mixing study that remains prolonged. Excess phospholipid corrects the abnormality. What is the most likely cause?

- A. Lupus anticoagulant
- B. Factor VIII deficiency
- C. Hemophilia B
- D. Afibrinogenemia

Answer Key & Rationales

Use this section after completing the questions.

1. B

Rationale: Tissue factor is exposed after vascular injury and binds factor VII/VIIa. The resulting complex helps activate factors IX and X, initiating thrombin generation.

2. B

Rationale: Factor XII participates in the contact pathway measured by the APTT in vitro. Its deficiency can markedly prolong the APTT even though it does not usually produce a bleeding tendency.

3. C

Rationale: Sodium citrate reversibly binds calcium and preserves coagulation factors well enough for routine clot-based testing.

4. C

Rationale: Routine coagulation specimens require approximately nine parts blood to one part citrate. Incorrect filling changes the effective citrate concentration and can distort clotting times.

5. B

Rationale: Factor X is part of the common pathway evaluated by both PT and APTT, so a significant deficiency can prolong both assays.

6. A

Rationale: Factor XIII stabilizes an already formed fibrin clot by cross-linking fibrin. PT and APTT measure events leading to initial fibrin formation, so they do not assess factor XIII.

7. B

Rationale: Vitamin K is needed for gamma-carboxylation of vitamin K-dependent proteins, including factors II, VII, IX, and X.

8. A

Rationale: D-dimer is produced when plasmin degrades fibrin that has been cross-linked by factor XIIIa, making it a marker of clot formation followed by fibrinolysis.

9. B

Rationale: The APTT assesses factors of the intrinsic pathway, including XII, XI, IX, and VIII, plus the common pathway factors.

10. B

Rationale: Underfilling leaves too much citrate relative to plasma. During testing, excess citrate can bind added calcium and artifactually prolong clotting times.

11. A

Rationale: Tissue plasminogen activator is produced by the body and promotes conversion of plasminogen to plasmin.

12. C

Rationale: Alpha-2 antiplasmin rapidly binds and inhibits free plasmin, limiting excessive fibrinolysis.

13. B

Rationale: TTP is characterized by platelet consumption in microvascular thrombi and MAHA. Unlike DIC, routine PT and APTT are generally normal.

14. A

Rationale: Aspirin irreversibly inhibits platelet cyclooxygenase, decreasing thromboxane A₂ production and thereby reducing platelet aggregation.

15. B

Rationale: GPIb serves as an important platelet receptor for VWF and is essential for platelet adhesion at sites of vascular injury.

16. B

Rationale: Platelet number is generally normal in von Willebrand disease. The disorder primarily affects VWF-mediated adhesion and can also reduce factor VIII activity.

17. B

Rationale: Bernard-Soulier syndrome results from a GPIb-related adhesion defect and classically features giant platelets with thrombocytopenia.

18. B

Rationale: Ristocetin-dependent agglutination requires the VWF-GPIb interaction. Bernard-Soulier syndrome lacks functional GPIb and therefore shows an abnormal ristocetin response.

19. A

Rationale: Glanzmann thrombasthenia is caused by a GPIIb/IIIa defect. Ristocetin-dependent adhesion remains intact, while aggregation to several other agonists is impaired.

20. A

Rationale: Acute immune thrombocytopenia commonly follows viral infection in children and presents with isolated thrombocytopenia and mucocutaneous bleeding.

21. B

Rationale: HUS is strongly associated with Shiga toxin-producing *E. coli*, particularly O157:H7, and may cause thrombocytopenia, MAHA, and renal injury.

22. A

Rationale: Storage pool deficiencies are platelet granule disorders, often involving reduced dense-granule contents and impaired secretion.

23. A

Rationale: Acquired TTP commonly results from autoantibodies against ADAMTS13, allowing unusually large VWF multimers to persist and promote platelet-rich microthrombi.

24. B

Rationale: Prostacyclin released by endothelium inhibits platelet aggregation, opposing the proaggregatory effect of thromboxane A₂.

25. A

Rationale: Hemophilia A is factor VIII deficiency. Because factor VIII belongs to the intrinsic pathway, APTT is prolonged while PT and platelet studies remain normal.

26. C

Rationale: Factor X is part of the common pathway, so its deficiency can prolong both PT and APTT. Thrombin time evaluates conversion of fibrinogen to fibrin and may remain normal.

27. A

Rationale: Vitamin K is fat soluble and requires bile for intestinal absorption. Obstructive jaundice can therefore produce vitamin K deficiency and impaired synthesis of functional vitamin K-dependent factors.

28. B

Rationale: Specific factor assays measure the patient's ability to correct plasma deficient in the factor being tested. Factor VIII-deficient plasma is therefore used to assess factor VIII activity.

29. A

Rationale: DIC consumes platelets and coagulation factors during widespread thrombin and fibrin generation, often prolonging both PT and APTT.

30. A

Rationale: Some malignancies, including adenocarcinomas, can release procoagulant substances and trigger disseminated coagulation.

31. A

Rationale: Factor XII strongly affects contact activation in laboratory APTT testing, yet physiologic clotting can proceed through tissue factor-dependent mechanisms.

32. A

Rationale: VWF supports platelet adhesion and carries factor VIII. VWF deficiency can therefore cause an abnormal ristocetin response and a prolonged APTT from reduced factor VIII.

33. A

Rationale: Severe factor VIII deficiency can cause spontaneous deep-tissue bleeding, particularly recurrent bleeding into joints.

34. C

Rationale: Factor XIII stabilizes fibrin after the initial clot forms. Its deficiency can cause delayed bleeding and poor wound healing while routine PT and APTT remain normal.

35. A

Rationale: Type 1 is the most common form and generally reflects a partial quantitative deficiency of VWF.

36. B

Rationale: Antithrombin is a major physiologic anticoagulant, and heparin greatly enhances its inhibitory activity against several activated coagulation enzymes.

37. B

Rationale: Normal plasma supplies missing factors and typically corrects a factor deficiency. Failure to correct suggests that an inhibitor remains active in the mixture.

38. A

Rationale: Lupus anticoagulants target phospholipid-associated reactions and can prolong phospholipid-dependent clotting tests such as APTT and DRVVT.

39. B

Rationale: DRVVT is phospholipid dependent and is commonly used as part of lupus anticoagulant evaluation.

40. A

Rationale: The INR standardizes PT results across thromboplastin reagents and is the routine laboratory measure used to monitor warfarin.

41. A

Rationale: Activated protein C downregulates coagulation by proteolytically inactivating factors Va and VIIIa, with protein S enhancing this action.

42. A

Rationale: Factor V Leiden makes factor Va relatively resistant to inactivation by activated protein C, increasing thrombotic risk.

43. B

Rationale: Prasugrel is a P2Y₁₂ inhibitor and reduces ADP-mediated platelet aggregation. Clopidogrel acts through the same receptor.

44. B

Rationale: The anti-factor Xa assay estimates heparin activity through inhibition of a known amount of factor Xa in an antithrombin-dependent system.

45. A

Rationale: Immune HIT results from antibodies against complexes of heparin and platelet factor 4. These immune complexes activate platelets and can cause both thrombocytopenia and thrombosis.

46. B

Rationale: Argatroban directly inhibits thrombin and can be used when heparin must be stopped because of HIT.

47. A

Rationale: The Bethesda assay measures the inhibitory effect of patient plasma on factor VIII activity and reports the inhibitor titer.

48. A

Rationale: The mutation makes activated factor V resistant to cleavage by activated protein C, allowing procoagulant activity to persist longer.

49. A

Rationale: Aspirin affects platelet function rather than platelet number or routine coagulation pathways, so platelet aggregation testing can demonstrate its effect.

50. B

Rationale: A recent viral illness followed by abrupt isolated thrombocytopenia and mucocutaneous bleeding in a child strongly supports acute immune thrombocytopenia.

51. B

Rationale: The combination of mucocutaneous bleeding, reduced VWF measurements, abnormal ristocetin response, and reduced factor VIII-related activity is characteristic of von Willebrand disease.

52. A

Rationale: Cryoprecipitate is rich in fibrinogen and is useful when substantial fibrinogen replacement is required.

53. A

Rationale: HIT should be suspected when thrombosis occurs with a substantial platelet-count decline during heparin exposure.

54. A

Rationale: Factor VIII inhibitors may show time- and temperature-dependent inhibition. Immediate correction followed by loss of correction after incubation is a classic clue.

55. A

Rationale: A markedly abnormal PT/INR signals increased procedural bleeding risk. The coagulation abnormality should be evaluated and managed before an invasive procedure.

56. A

Rationale: Preanalytical problems can consume coagulation factors and produce unexpected results. Inspecting the specimen for clotting is an appropriate first troubleshooting step.

57. A

Rationale: Prolonged refrigeration can cause cold activation of factor VII and artifactually shorten PT.

58. A

Rationale: Heparin depends heavily on antithrombin for its anticoagulant effect. Antithrombin deficiency can therefore produce a poor response to heparin.

59. A

Rationale: A major platelet decline plus new thrombosis during heparin exposure creates strong clinical concern for HIT. A single insensitive or negative platelet aggregation result should not override a convincing clinical picture.

60. A

Rationale: Failure to correct on mixing suggests an inhibitor, while correction with excess phospholipid supports a phospholipid-dependent inhibitor such as lupus anticoagulant.